

Ventricular system and subarachnoid space

CHAPTER 18

The cerebral ventricular system consists of a series of interconnecting spaces and channels within the brain (Figs 18.1 and 18.2), which are derived from the central lumen of the embryonic neural tube and the cerebral vesicles to which it gives rise (Ch. 17). Each cerebral hemisphere contains a large lateral ventricle that communicates near its rostral end with the third ventricle via the interventricular foramen (foramen of Monro). The third ventricle is a midline, slit-like cavity lying between the right and left thalamus and hypothalamus. Caudally, the third ventricle is continuous with the cerebral aqueduct, a narrow tube that passes the length of the midbrain, and which is continuous in turn with the fourth ventricle, a wide cavity lying between the brainstem and cerebellum. The fourth ventricle communicates with the subarachnoid space of the cisterna magna through the foramen of Magendie, and with the cerebellopontine angles through the foramina of Luschka; caudally it is continuous with the vestigial central canal of the spinal cord.

The ventricular system contains cerebrospinal fluid (CSF), which is mostly secreted by the choroid plexuses located within the lateral, third and fourth ventricles.

TOPOGRAPHY AND RELATIONS OF THE VENTRICULAR SYSTEM

LATERAL VENTRICLE

Viewed from its lateral aspect, the lateral ventricle has a roughly C-shaped profile (see Figs 18.1–18.2; Fig. 18.3). The shape is a consequence of the developmental expansion of the frontal, parietal and occipital regions of the hemisphere, which displaces the temporal lobe inferiorly and anteriorly. Both the caudate nucleus and the fornix, which lie in the wall of the ventricle, have adopted a similar morphology, so that the tail of the caudate nucleus encircles the thalamus in a C shape, and the fornix traces the outline of the ventricle forwards to the interventricular foramen.

The lateral ventricle is customarily divided into a body and anterior (frontal), posterior (occipital) and inferior (temporal) horns (Rhoton 2002).

The anterior horn lies within the frontal lobe. The posterior aspect of the genu and the rostrum of the corpus callosum bound it anteriorly, and its roof is formed by the anterior part of the body of the corpus callosum. The anterior horns of the two ventricles are separated by the septum pellucidum. The coronal profile of the anterior horn is roughly that of a flattened triangle in which the rounded head of the caudate nucleus forms the lateral wall and floor (Figs 18.4–18.5). The anterior horn extends back as far as the interventricular foramen.

Cavum septum pellucidum (CSP) describes a septum pellucidum that has a separation between its two leaflets (septal laminae). The cavity contains cerebrospinal fluid that filters from the ventricles through the septal laminae. The most common type of cavum septum pellucidum is non-communicating, i.e. it is not connected to the ventricular system. Cavum septum pellucidum is present in 100% of fetuses below 36 weeks' gestation. Its prevalence at 36, 38 and 40 weeks is 69%, 54% and 36% respectively, decreasing to 10% in children under 1 year, 5–6% at 2–5 years, 2.7% at 6–9 years and 2.3% at 10–14 years (Farrugia 1981, Mott et al 1992, Nakano et al 1981).

The body of the lateral ventricle lies within the frontal and parietal lobes, and extends from the interventricular foramen to the splenium of the corpus callosum. The bodies of the lateral ventricles are separated by the septum pellucidum, which contains the columns of the fornices in its lower edge. The lateral wall of the body of the ventricle is formed by the caudate nucleus superiorly and the thalamus inferiorly. The boundary between the thalamus and caudate nucleus is marked by a groove that is occupied by a fascicle of nerve fibres, the stria terminalis, and by the superior thalamostriate vein (see Fig. 18.4). The inferior limit of the body of the ventricle and its medial wall are formed by the body of the fornix (Fig. 18.6). The fornix is separated from the thalamus by the choroidal fissure. The choroid plexus occludes the choroidal fissure and covers part of the thalamus and fornix (Nagata 1988) (Fig 18.7). The body of the lateral ventricle widens posteriorly to become

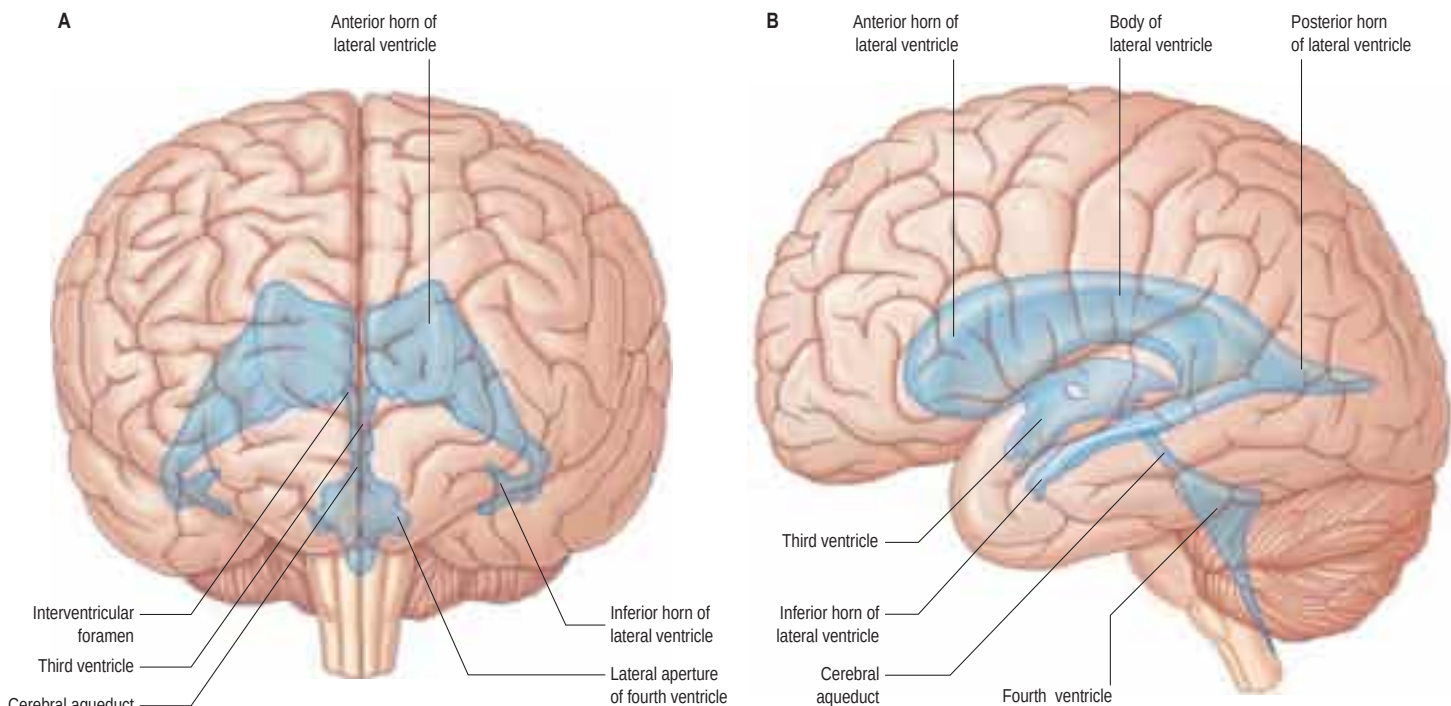


Fig. 18.1 The ventricular system. A, Anterior view. B, Left lateral view.

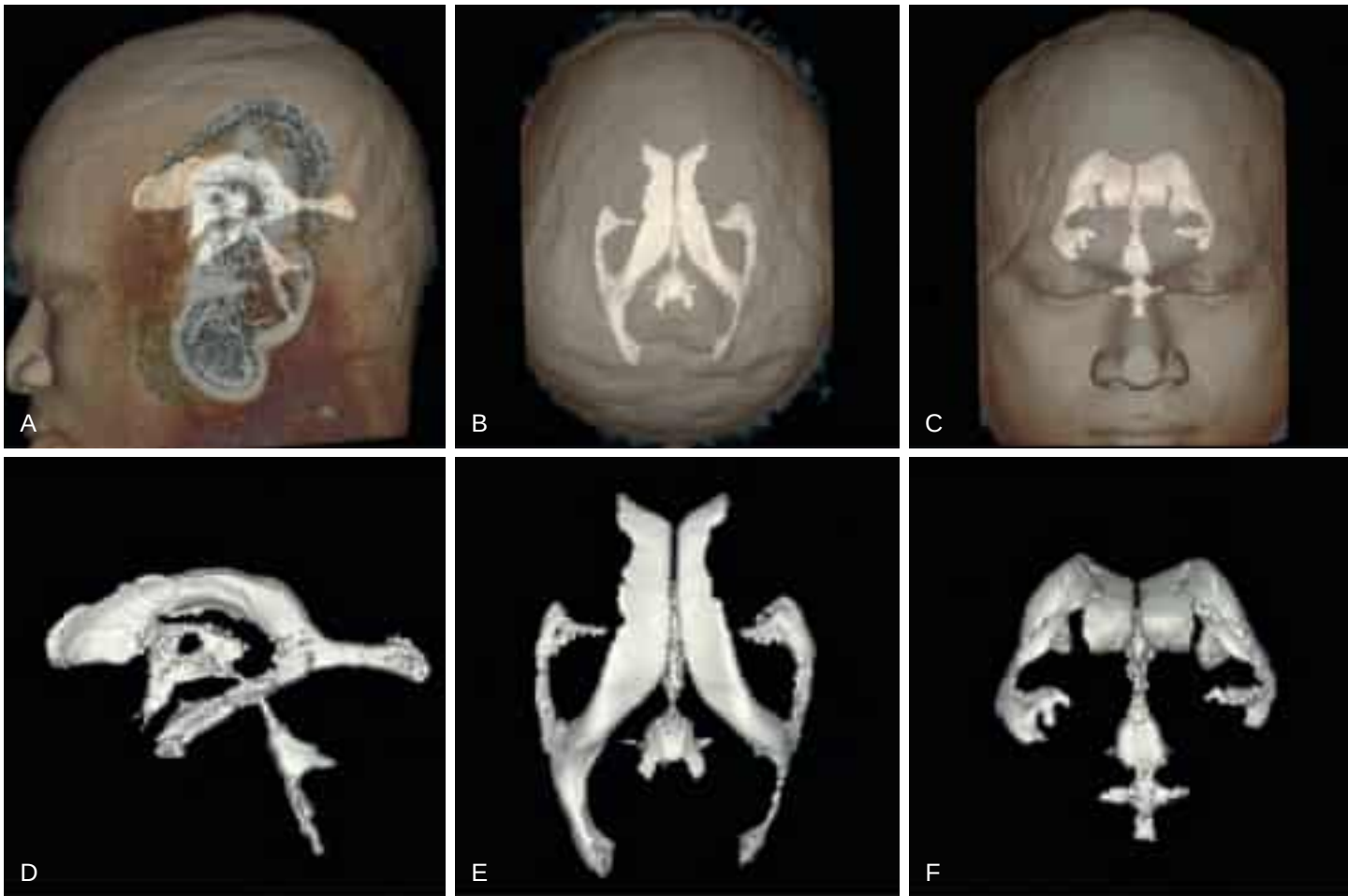


Fig. 18.2 The ventricular system. Three-dimensional computer-reformatted volumetric T2-weighted magnetic resonance images display the ventricles in lateral (**A, D**), superior (**B, E**) and anterior (**C, F**) views. **A–C**, Orientation of the ventricular system within the head, seen through the partially transparent skin surface. **D–F**, Magnified display of the isolated ventricles in the same orientation. See also Videos 18.1 and 18.2, and 28.1–28.3 for an interactive rotation. (With permission from Rios JC, Galper MW, Naidich TP, Ventricles and intracranial subarachnoid spaces. In: Naidich TP, Castillo M, Cha S, Smirniotopoulos JG (eds), *Imaging of the Brain*. 2013, Saunders, Elsevier.)

continuous with the posterior and inferior horns at the collateral trigone or atrium.

The posterior horn curves posteromedially into the occipital lobe. It is usually diamond-shaped or square in outline, and the two sides are often asymmetrical. Fibres of the tapetum of the corpus callosum separate the ventricle from the optic radiation, and form the roof and lateral wall of the posterior horn. Fibres of the splenium of the corpus callosum (forceps major) pass medially as they sweep back into the occipital lobe, and produce a rounded elevation in the upper medial wall of the posterior horn. Lower down, a second elevation, the calcar avis, corresponds to the deeply infolded cortex of the anterior part of the calcarine sulcus.

The inferior horn is the largest compartment of the lateral ventricle and extends forwards into the temporal lobe. It curves round the posterior aspect of the thalamus (pulvinar), passes downwards and posterolaterally and then curves anteriorly to end within 2.5 cm of the temporal pole, near the uncus. Its position relative to the surface of the hemisphere usually corresponds to the superior temporal sulcus. The roof of the inferior horn is formed mainly by the tapetum of the corpus callosum, but also by the tail of the caudate nucleus and the stria terminalis, which extend forwards in the roof to terminate in the amygdala at the anterior end of the ventricle. The floor of the inferior horn consists of the hippocampus medially and the collateral eminence, formed by the infolding of the collateral sulcus, laterally. The inferior part of the choroid fissure lies between the fimbria (a distinct bundle of efferent fibres that leaves the hippocampus) and the stria terminalis in the roof of the inferior horn (Fig. 18.8). The temporal extension of the choroid plexus fills this fissure and covers the outer surface of the hippocampus.



Fig. 18.3 A T1-weighted sagittal MRI, showing the C-shaped profile of the lateral ventricle.

THIRD VENTRICLE

The third ventricle is a midline, slit-like cavity, which is derived from the primitive forebrain vesicle (see Figs 18.1, 18.6, 15.6; Figs 18.9–18.10) (Rhoton 2002). The upper part of the lateral wall of the ventricle is formed by the medial surface of the anterior two-thirds of the thalamus, and the lower part is formed by the hypothalamus anteriorly and the subthalamus posteriorly. An indistinct hypothalamic sulcus extends horizontally on the ventricular wall between the interventricular foramen and the cerebral aqueduct, and marks the boundary between the thalamus and hypothalamus. Dorsally, the lateral wall is limited by a ridge covering the stria medullaris thalami. The lateral walls of the third ventricle are commonly joined by an interthalamic adhesion, or massa intermedia, a band of grey matter that extends from one thalamus to the other. An interthalamic adhesion is almost always found in humans, more often in women, in whom it is larger by an average of 53% (Orly 2005).

Anteriorly, the third ventricle extends to the lamina terminalis (see Fig. 18.10). This thin structure stretches from the optic chiasma to the rostrum of the corpus callosum and represents the rostral boundary of the embryonic neural tube. The lamina terminalis forms the roof of a small virtual cavity, the cistern of the lamina terminalis, which lies immediately below the ventricle and is an extension of the interpeduncular cistern. This is clinically important because it contains the anterior communicating artery; aneurysm development and rupture at this site may cause intraventricular haemorrhage through the thin membrane of the lamina terminalis. Above the lamina terminalis, the anterior wall of the third ventricle is formed by the diverging columns of the fornices and the transversely orientated anterior commissure, which crosses the midline. The narrow interventricular foramen is located immediately posterior to the column of the fornix and separates the fornix from the anterior nucleus of the thalamus.



Fig. 18.5 A coronal T2-weighted MRI scan, at the level of the anterior horn of the lateral ventricle.

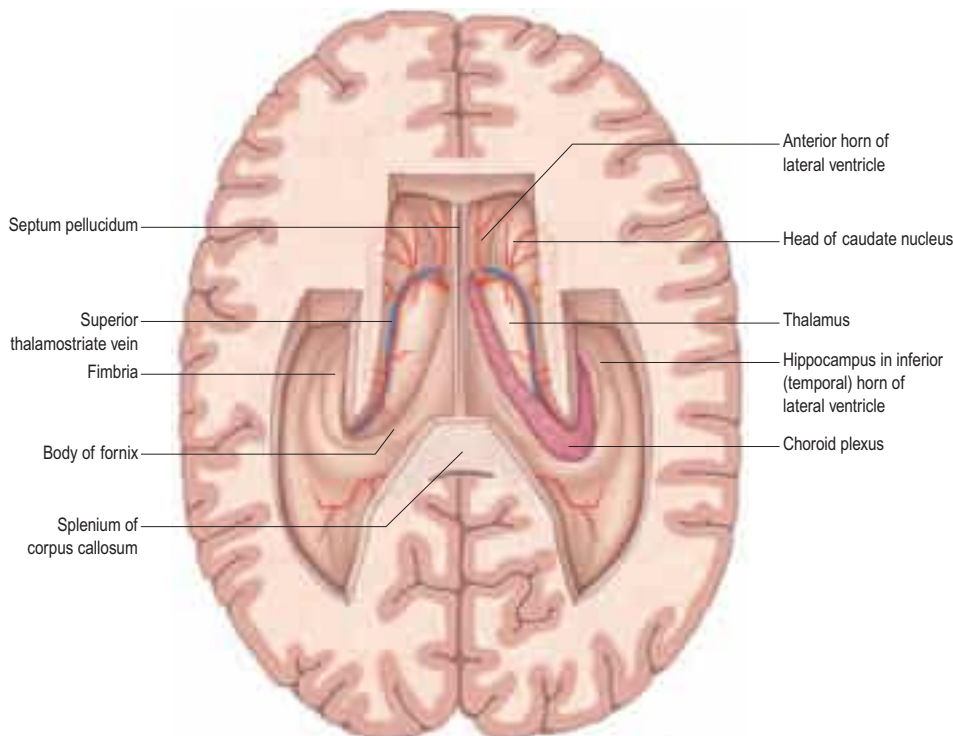


Fig. 18.4 A horizontal section of the cerebrum dissected to remove the roofs of the lateral ventricles.

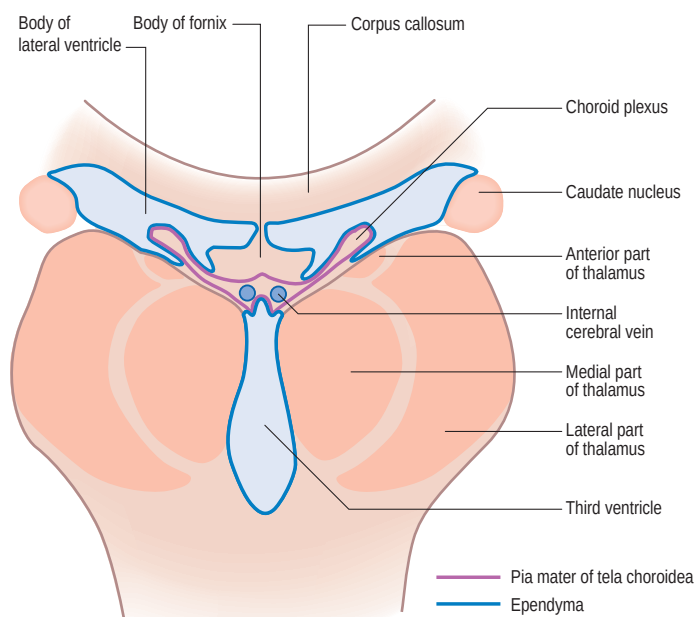


Fig. 18.6 The lateral and third ventricles, coronal section.

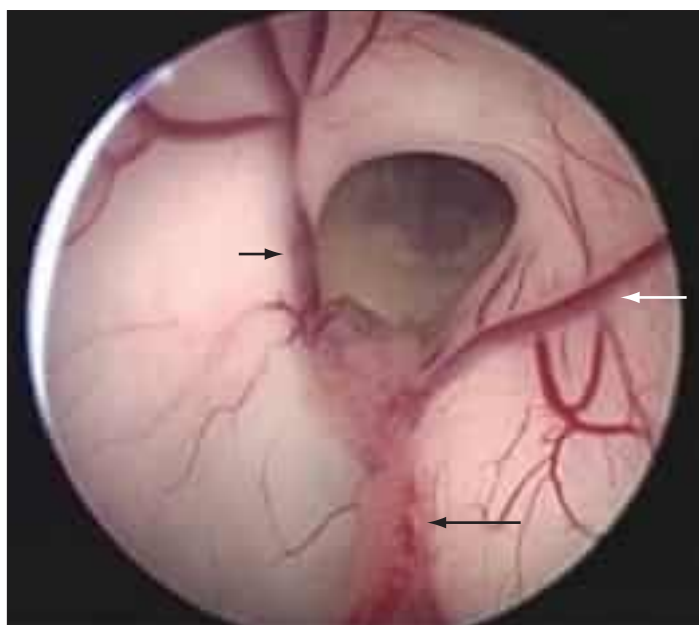


Fig. 18.7 An endoscopic view into the right lateral ventricle. The thalamostriate vein (white arrow) runs along the lateral wall and converges with the anterior septal vein (short black arrow) before converging and entering the foramen of Monro as the internal cerebral vein. The superior choroidal vein (long black arrow) is also seen coursing through the choroid plexus. (With permission from Winn HR, editor. *Youmans Neurological Surgery*. 6th edition. 2011 Philadelphia, W. B. Saunders, Elsevier.)

A small, angular, optic recess occurs inconsistently at the base of the lamina terminalis, just dorsal to and extending into the optic chiasma. Behind it, the anterior part of the floor of the third ventricle is formed mainly by hypothalamic structures. The thin infundibular recess lies immediately behind the optic chiasma and extends into the pituitary stalk. Behind this recess, the tuber cinereum and the mammillary bodies form the floor of the ventricle (see Fig. 18.10; Fig. 18.11).

The roof of the third ventricle is a thin ependymal layer that extends from its lateral walls to the choroid plexus, which spans the choroidal fissure. The body of the fornix lies above the roof (see Fig. 18.6).

The posterior boundary of the ventricle is marked by a suprapineal recess, the habenular commissure, a pineal (epiphysial) recess, which extends into the pineal stalk, and by the posterior commissure. Below the posterior commissure the ventricle is continuous with the cerebral aqueduct of the midbrain (see Fig. 18.10).

The third ventricle is larger in infants with trisomy 21 compared with controls (Schimmel et al 2006).

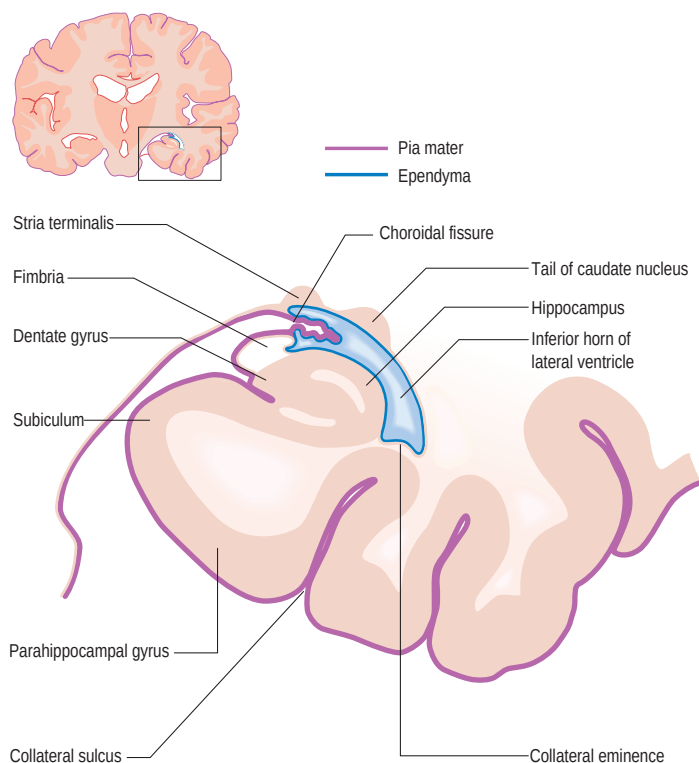


Fig. 18.8 The inferior horn of the left lateral ventricle, coronal section.

CEREBRAL AQUEDUCT

The cerebral aqueduct is a small tube, roughly circular in transverse section and 1–2 mm in diameter. The length of the aqueduct in children, as measured at necropsy, is shown in Table 18.1. In children, the mean cross-sectional area of the aqueduct is 0.5 mm², and the narrowest cross-sectional area compatible with normal ventricular size is 0.1 mm² (Emery and Staschak 1972).

The aqueduct extends throughout the dorsal quarter of the midbrain in the midline and is surrounded by the periaqueductal (central) grey matter (see Figs 18.10, 21.17). Rostrally, it commences immediately below the posterior commissure, where it is continuous with the caudal aspect of the third ventricle. Caudally, it is continuous with the lumen of the fourth ventricle at the junction of the midbrain and pons. The superior and inferior colliculi are dorsal to the aqueduct and the midbrain tegmentum is ventral. Primary aqueductal stenosis is a feature of von Recklinghausen's disease in children (Spadaro et al 1986); this is not the only cause and population studies suggest that genetic factors may play a more important role in hydrocephalus than previously accepted (Munch et al 2012).

FOURTH VENTRICLE

The fourth ventricle lies between the brainstem and the cerebellum (see Figs 18.1, 18.9–18.10, 21.5; Fig. 18.12) (Rhoton 2000). Rostrally, it is continuous with the cerebral aqueduct, and caudally with the central canal of the spinal cord. In sagittal section, the fourth ventricle has a characteristic triangular profile, and the apex of its tented roof protrudes into the inferior aspect of the cerebellum. The ventricle is at its widest at the level of the pontomedullary junction, where a lateral recess on both sides extends to the lateral border of the brainstem. At this point the lateral apertures of the fourth ventricle (foramina of Luschka) open into the subarachnoid space at the cerebellopontine angle, behind the upper roots of the glossopharyngeal nerves.

The floor of the fourth ventricle is a shallow diamond-shaped, or rhomboidal, depression (rhomboid fossa) on the dorsal surfaces of the pons and the rostral half of the medulla (see Fig. 18.2). It consists largely of grey matter and contains important cranial nerve nuclei (V–XII). The precise location of some nuclei is discernible from surface features. The superior part of the ventricular floor is triangular in shape and is limited laterally by the superior cerebellar peduncles as they converge towards the cerebral aqueduct. The inferior part of the ventricular floor is also triangular in shape and is bounded caudally by the gracile and cuneate tubercles, which contain the dorsal column nuclei, and, more rostrally, by the diverging inferior cerebellar peduncles. A longitudinal median sulcus divides the floor of the fourth ventricle.

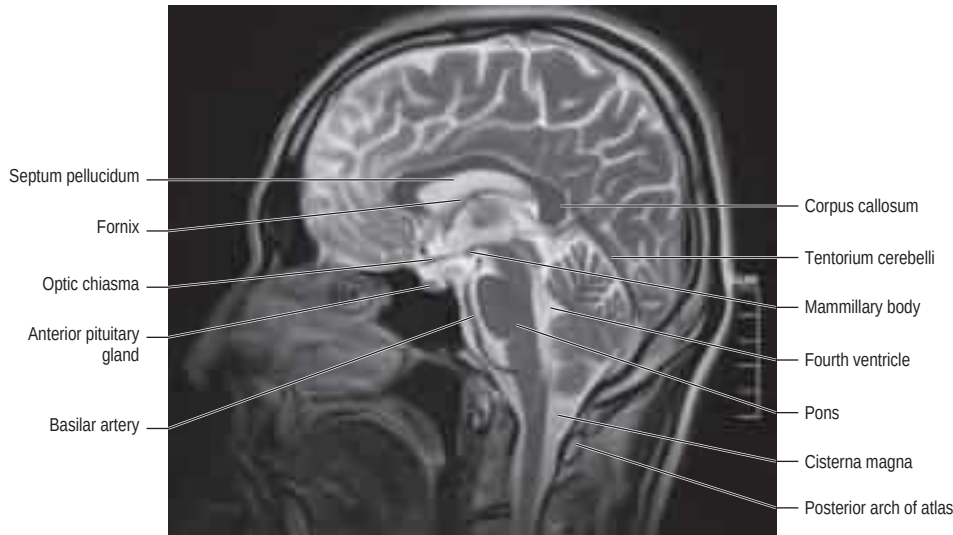


Fig. 18.9 A sagittal T2-weighted MRI scan of the head.

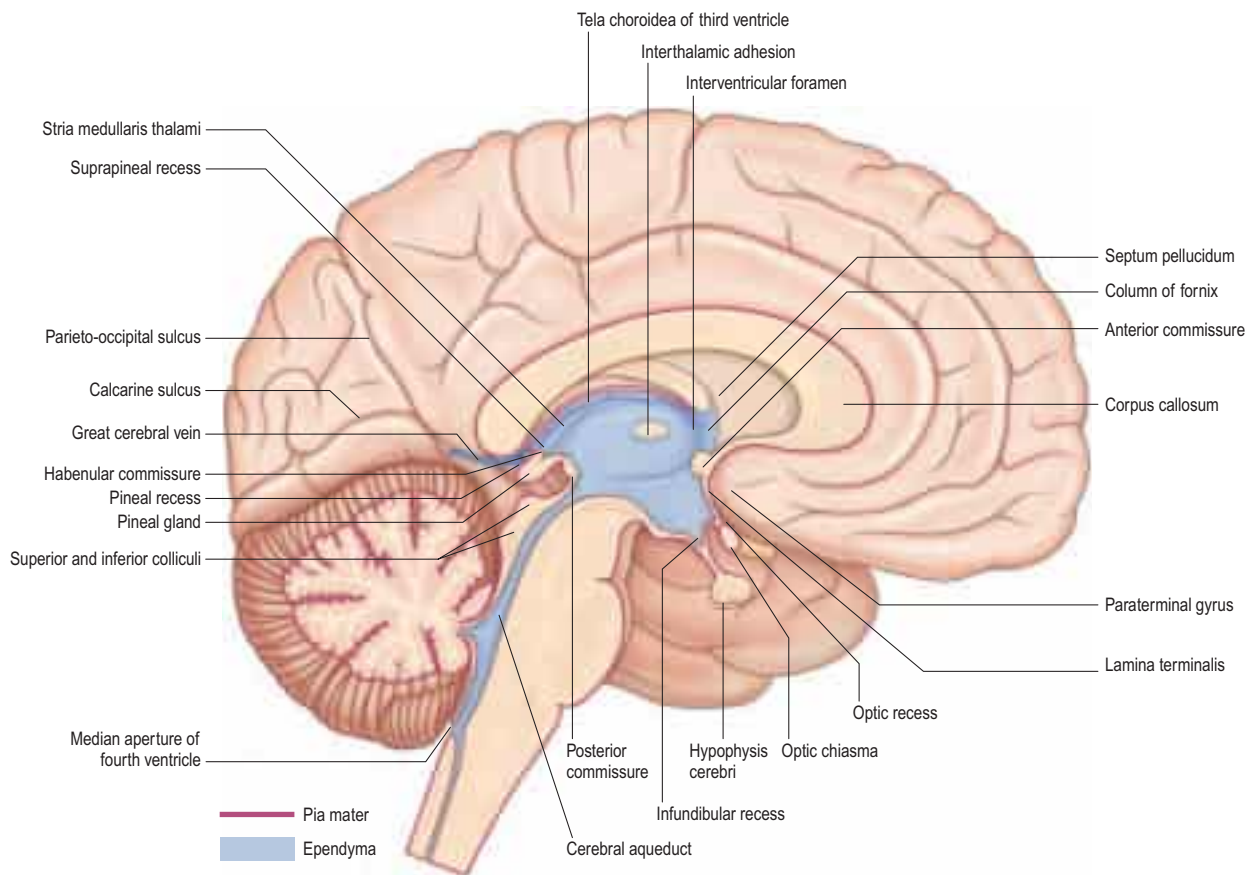


Fig. 18.10 The third and fourth ventricles, sagittal hemisection of the brain.

Each half is itself divided, by an often indistinct sulcus limitans, into a medial region known as the medial eminence and a lateral region known as the vestibular area. The vestibular nuclei lie beneath the vestibular area. In the superior part, the medial eminence is represented by the facial colliculus, a small elevation produced by an underlying loop of efferent fibres from the facial nucleus, which covers the abducens nucleus. Between the facial colliculus and the vestibular area the sulcus limitans widens into a small depression, the superior fovea. In its upper part, the sulcus limitans constitutes the lateral limit of the floor of the fourth ventricle. Here, a small region of bluish-grey pigmentation denotes the presence of the subjacent locus coeruleus. Caudal to the facial colliculus, at the level of the lateral recess of the ventricle, a variable group of nerve fibre fascicles, known as the striae medullaris, runs transversely across the ventricular floor and passes into the median sulcus. In the inferior part, the medial eminence is represented by the hypoglossal triangle (trigone), which lies over the hypoglossal nucleus. Laterally, the sulcus limitans widens to produce an indistinct inferior fovea. Caudal to the inferior fovea, between the hypoglossal triangle

and the vestibular area, is the vagal triangle (trigone), which covers the dorsal motor nucleus of the vagus. A narrow translucent ridge, the funiculus separans, which is separated from the gracile tubercle by the small area postrema, crosses below the vagal triangle.

The roof of the fourth ventricle is formed by the superior and inferior medullary veli. The thin superior medullary velum stretches across the ventricle between the converging superior cerebellar peduncles (see Fig. 18.12) and is continuous with the cerebellar white matter. Dorsally, it is covered by the lingula of the superior vermis. The inferior medullary velum is more complex and is mostly composed of a thin sheet, devoid of neural tissue, formed by ventricular ependyma and the pia mater of the tela choroidea. Just inferior to the nodule of the cerebellum, a median aperture, the foramen of Magendie, opens the roof of the fourth ventricle into the cisterna magna.

The aperture forms when a membranous structure (Blake's pouch) perforates into the fourth ventricle at the ninth week of fetal development; persistence of this membrane results in cystic obstruction of the median outlet from the fourth ventricle (Paladini et al 2012).

CIRCUMVENTRICULAR ORGANS

The walls of the ventricular system are lined with ependymal cells that cover a subependymal layer of glia. At certain midline sites in the ventricular walls, collectively referred to as circumventricular organs, the blood–brain barrier is absent and specialized ependymal cells called

tanycytes are present (Horsburg and Massoud 2013) (Fig. 18.13). The functions of ependyma and tanycytes may include secretion into the CSF; transport of neurochemicals from subjacent neurones, glia or vessels to the CSF; transport of neurochemicals from the CSF to the same subjacent structures; and chemoreception (Rodriguez et al 2005). Adult mammalian neurogenesis occurs in discrete neurogenic niches that are best characterized in the ependymal and subependymal glial cell layers in the subgranular zone of the dentate gyrus and in the subventricular zone; the existence of adult human neurogenic niches is controversial.

The circumventricular organs include the vascular organ (organum vasculosum), subfornical organ, neurohypophysis, median eminence, subcommissural organ, pineal gland and area postrema (see Fig. 18.13).

The vascular organ lies in the lamina terminalis between the optic chiasma and the anterior commissure. Its external zone contains a richly fenestrated vascular plexus that covers glia and a network of nerve fibres. The ependymal cells of the vascular organ, like those of other circumventricular organs, are flattened and have few cilia. The major inputs appear to come from the subfornical organ, locus coeruleus and a number of hypothalamic nuclei. The vascular organ projects to the median preoptic and supraoptic nuclei. It is involved in the regulation of fluid balance and may also have neuroendocrine functions.

The subfornical organ lies at the level of the interventricular foramen. It contains many neurones, glial cells and a dense fenestrated capillary

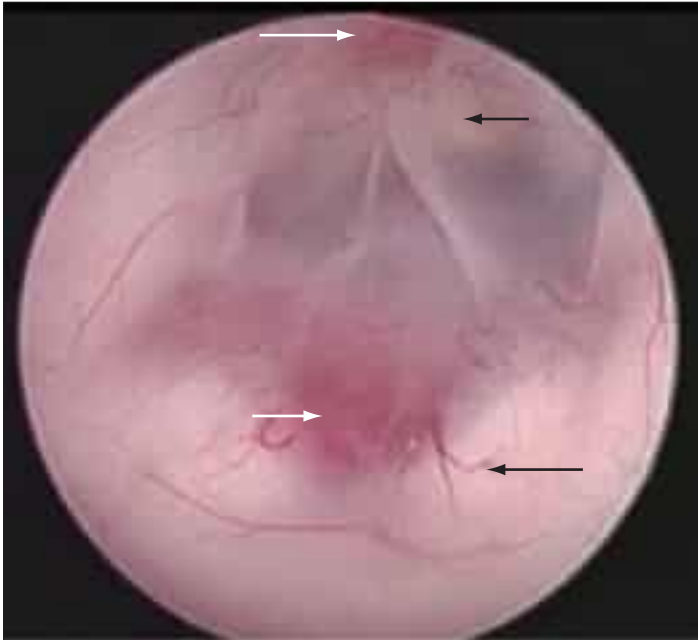


Fig. 18.11 An endoscopic view of the floor of the third ventricle. The ideal spot for fenestration is in the midline, midway between the dorsum sellae (short black arrow) and the basilar artery (short white arrow). The paired mammillary bodies (long black arrow) and the infundibular recess (long white arrow) are also seen. (With permission from Winn HR. Youmans Neurological Surgery. 6th edition. 2011 Philadelphia, W. B. Saunders, Elsevier.)

Table 18.1 The length of the cerebral aqueduct, as measured at necropsy in children

Crown–rump length (cm)	Length of aqueduct (mm)	Apparent equivalent age of children	Mean range
35	12.8	9.5–15.0	Birth to 6 weeks
45	15.5	13.0–17.6	3–9 months
55	17.4	15.2–18.8	1–4 years
65	18.2	16.6–19.4	5–7 years
75	18.8	17.6–20.0	≥8 years

(With permission from Emery JL, Staschak MC. The size and form of the cerebral aqueduct in children. Brain 1972; 95:591–598.)

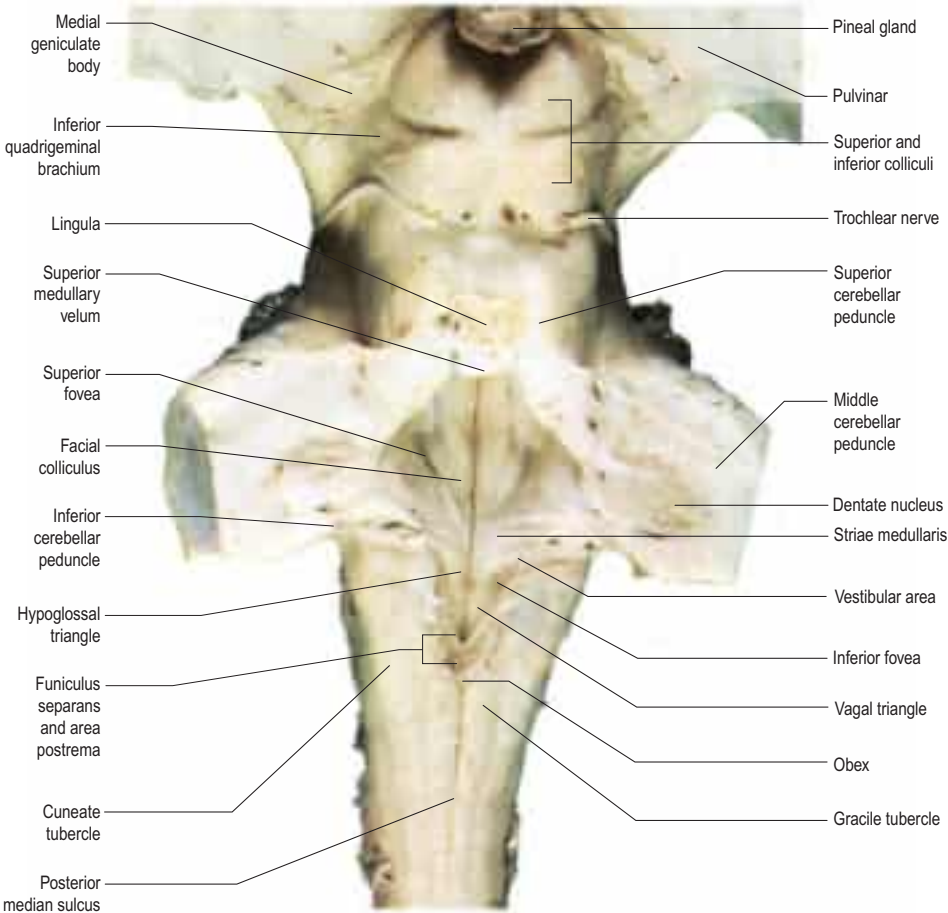


Fig. 18.12 The floor of the fourth ventricle, dorsal aspect of the brainstem.

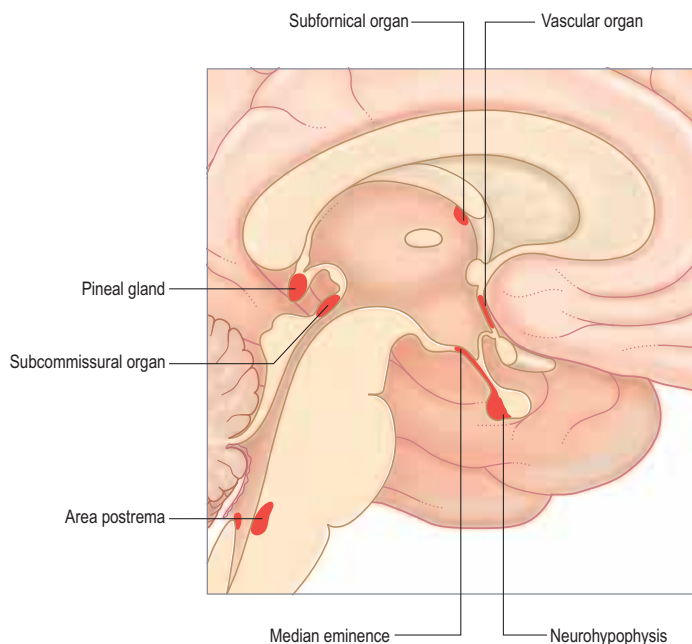


Fig. 18.13 The locations of the circumventricular organs, median sagittal section of the brain.

plexus, and is covered by flattened ependyma. It is believed to have widespread hypothalamic interconnections and to function in the regulation of fluid balance and thirst.

The neurohypophysis (posterior pituitary) is the site of termination of neurosecretory projections from the supraoptic and paraventricular nuclei of the hypothalamus. Neurones in these nuclei release vasopressin and oxytocin respectively into the capillary bed of the neurohypophysis, where the hormones gain access to the general circulation.

The median eminence contains the terminations of axons of hypothalamic neurosecretory cells. Peptides released from these axons control the hormonal secretions of the anterior pituitary via the pituitary portal system of vessels.

The subcommissural organ lies ventral to and below the posterior commissure, near the inferior wall of the pineal recess.

The pineal gland is a small structure, approximately 8 mm in diameter, situated rostradorsal to the superior colliculus and behind the stria medullaris.

The area postrema is a bilaterally paired structure, located at the caudal limit of the floor of the fourth ventricle. It is an important chemoreceptive area that triggers vomiting in response to the presence of emetic substances in the blood. The area postrema, along with the nucleus of the solitary tract and the dorsal motor nucleus of the vagus, makes up the so-called dorsal vagal complex, which is the major termination site of vagal afferent nerve fibres.

CHOROID PLEXUS AND CEREBROSPINAL FLUID

CHOROID PLEXUS

The vascular pia mater in the roofs of the third and fourth ventricles, and in the medial wall of the lateral ventricle along the line of the choroid fissure, is closely apposed to the ependymal lining of the ventricles, without any intervening brain tissue. It forms the tela choroidea, which gives rise to the highly vascularized choroid plexuses from which CSF is secreted into the lateral, third and fourth ventricles (see Figs 18.4, 18.6, 18.8). The body or stroma of the choroid plexus consists of many capillaries, separated from the ventricles by the pia mater and choroid ependymal cells (Strazielle 2000).

In the lateral ventricle, the choroid plexus extends anteriorly as far as the interventricular foramen, through which it is continuous across the third ventricle with the plexus of the opposite lateral ventricle. From the interventricular foramen, the plexus passes posteriorly, in contact with the thalamus, curving round its posterior aspect to enter the inferior horn of the ventricle and reach the hippocampus. Throughout the body of the ventricle, the choroid fissure lies between the fornix superiorly and the thalamus inferiorly (see Fig. 18.6).

From above, the tela choroidea is triangular with a rounded apex between the interventricular foramina, often indented by the anterior

columns of the fornices. Its lateral edges are irregular and contain choroid vascular fringes. At the posterior basal angles of the tela, these fringes continue and curve on into the inferior horn of the ventricle. When the tela is removed, a transverse slit (the transverse fissure) is left between the splenium and the junction of the ventricular roof with the tectum. The transverse fissure contains the roots of the choroid plexus of the third ventricle and of the lateral ventricles.

The choroid plexus of the third ventricle is attached to the tela choroidea, which is, in effect, the thin roof of the third ventricle as it develops during fetal life. In coronal sections of the cerebral hemispheres, it can be seen that the choroid plexus of the third and lateral ventricles are continuous.

The choroid plexus of the fourth ventricle is similar in structure to that of the lateral and third ventricles. The roof of the inferior part of the fourth ventricle develops as a thin sheet in which the pia mater is in direct contact with the ependymal lining of the ventricle. This thin sheet, the tela choroidea of the fourth ventricle, lies between the cerebellum and the inferior part of the roof of the ventricle. The choroid plexus of the fourth ventricle is T-shaped, with vertical and horizontal limbs, but the precise form varies widely from a single vertical limb to an elongated 'T' that extends out through the foramina of Luschka into the cerebellopontine angle. The vertical (longitudinal) limb is double, flanks the midline and is adherent to the roof of the ventricle. The limbs fuse at the superior margin of the median aperture (foramen of Magendie) and are often prolonged on to the ventral aspect of the cerebellar vermis. The horizontal limbs of the plexus project into the lateral recesses of the ventricle. Small tufts of plexus may pass through the lateral apertures (foramina of Luschka) and emerge, still covered by ependyma, in the subarachnoid space of the cerebellopontine angle. The mean thickness of the choroid plexus in the fourth ventricle in children is 2.5 mm (Madhukar et al 2012).

The blood supply of the choroid plexus in the tela choroidea of the lateral and third ventricles is usually via a single vessel from the anterior choroidal branch of the internal carotid artery and several choroidal branches of the posterior cerebral artery; the two sets of vessels anastomose to some extent. Capillaries drain into a rich venous plexus served by a single choroidal vein. The blood supply of the fourth ventricular choroid plexus is from the inferior cerebellar arteries.

Physiological calcification of the choroid plexus and pineal gland are the most frequently described intracranial calcifications incidentally discovered during head computed tomography (CT) examinations (Doyle and Anderson 2006).

CEREBROSPINAL FLUID

CSF is a clear, colourless liquid. Normal CSF contains small amounts of protein and differs from blood in its electrolyte content. It is not simply an ultrafiltrate of blood but is actively secreted by the choroid plexuses in the lateral, third and fourth ventricles.

The choroid plexus epithelium constitutes a blood-CSF barrier. Choroid plexus epithelial cells (see Figs 3.17, 3.18) have the characteristics of transport and secretory cells; their apical surfaces have microvilli, and their basal surfaces exhibit interdigitations and folding. Tight junctions (occluding junctions, zonulae adherentes) at the apical ends of the cells are permeable to small molecules. Fenestrated capillaries lie just beneath the epithelial cells in the stroma of the choroid plexus. The ependymal lining of the ventricles and the extracellular fluid from the brain parenchyma are additional sources of CSF, but how much each source contributes to CSF production is unclear.

SUBARACHNOID SPACE AND CIRCULATION OF CEREBROSPINAL FLUID

SUBARACHNOID SPACE

The subarachnoid space lies between the arachnoid and the pia mater (Videos 18.1 and 18.2). It is continuous with the lumen of the fourth ventricle via the median aperture (foramen of Magendie) and the paired lateral apertures (foramina of Luschka). The subarachnoid space contains CSF; the larger arteries and veins that traverse the surface of the brain, and the intracranial or intravertebral portions respectively of the cranial and spinal nerves.

Trabeculae, in the form of sheets or fine filiform structures, each containing a core of collagen, cross the subarachnoid space from the deep layers of the arachnoid mater to the pia mater (Fig. 18.14). The trabeculae are attached to the large blood vessels within the subarachnoid space and may form compartments, particularly in the perivascular

The ependymal cells on the dorsal aspect of the cerebral aqueduct are tall, columnar and ciliated, with granular basophilic cytoplasm; they may be involved in the secretion of materials into the CSF from adjoining axonal terminals or capillaries.

It is part of the epithalamus and consists of internal lobules of pinealocytes and sparse neurones covered by a pial capsule. It secretes melatonin, which appears to be involved in the inhibition of puberty in children and in the regulation of the sleep-wake cycle in both children and adults.

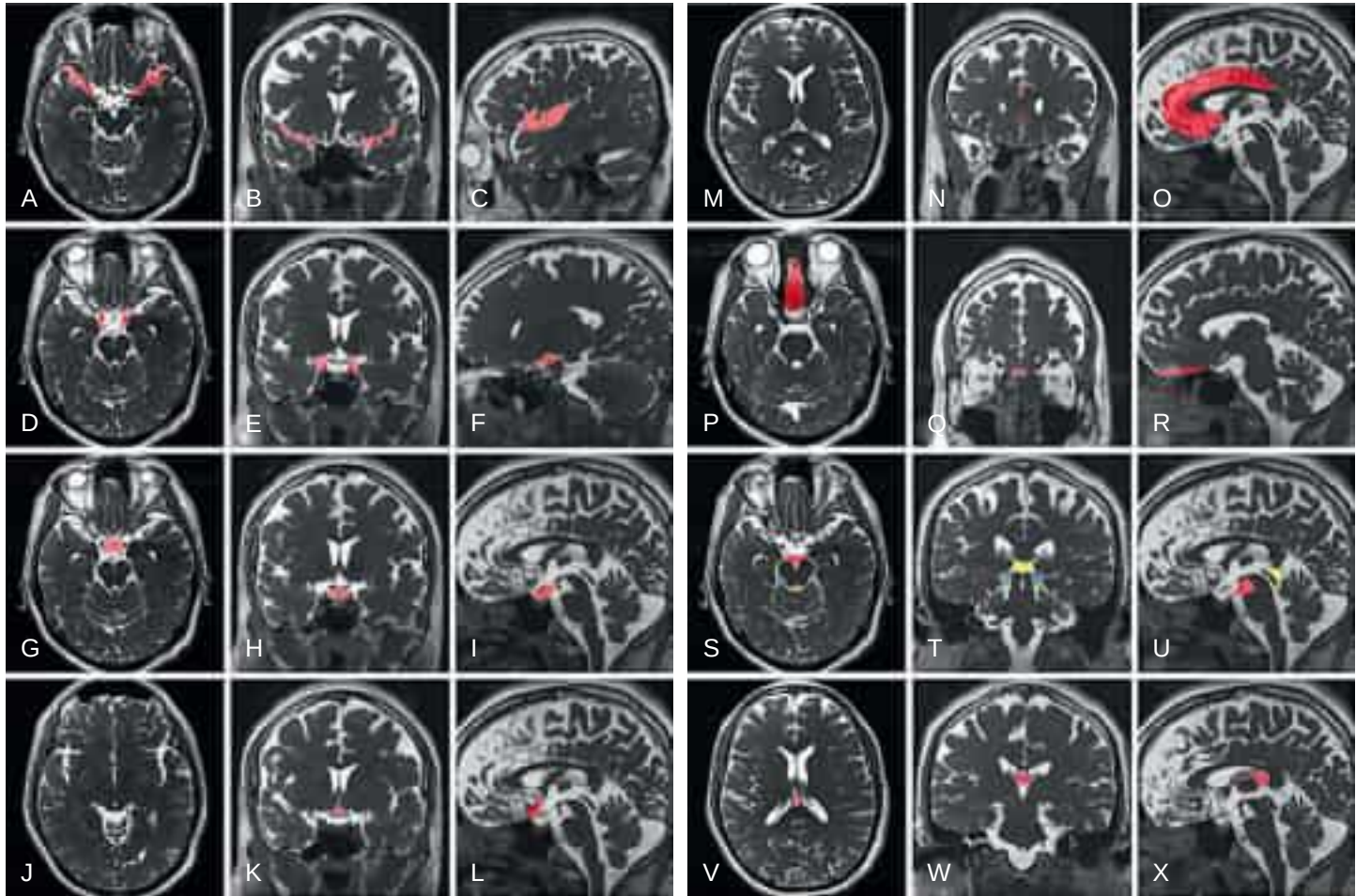


Fig. 18.16 The supratentorial subarachnoid cisterns. FIESTA MR images colour-coded to provide a visual approximation to the borders of the individually named supratentorial cisterns. These images are arrayed so that each row depicts a single cistern, in the axial plane (column 1), coronal plane (column 2) and sagittal plane (column 3). This montage is intended to indicate the general location for each cistern within the sea of multiple confluent CSF spaces around the brain. **A–C**, Paired Sylvian cisterns. **D–F**, Paired carotid cisterns. **G–I**, Midline chiasmatic cistern. **J–L**, Midline cistern of the lamina terminalis. **M–O**, Midline pericallosal cistern. **P–R**, Paired olfactory cisterns. **S–U**, Midline perimesencephalic cistern. **V–X**, Cistern of the velum interpositum. The perimesencephalic cistern is often considered in four portions (**S–U**): midventral interpeduncular fossa (red), paired anterolateral crural cisterns (green), paired posterolateral ambient cisterns (blue), and mid-dorsal quadrigeminal plate (vein of Galen) cistern (yellow). (With permission from Rios JC, Galper MW, Naidich TP. Ventricles and intracranial subarachnoid spaces. In: Naidich TP, Castillo M, Cha S, Smirniotopoulos JG (eds), *Imaging of the Brain*. Chapter 13, 245–271. 2013, Elsevier, Saunders.)

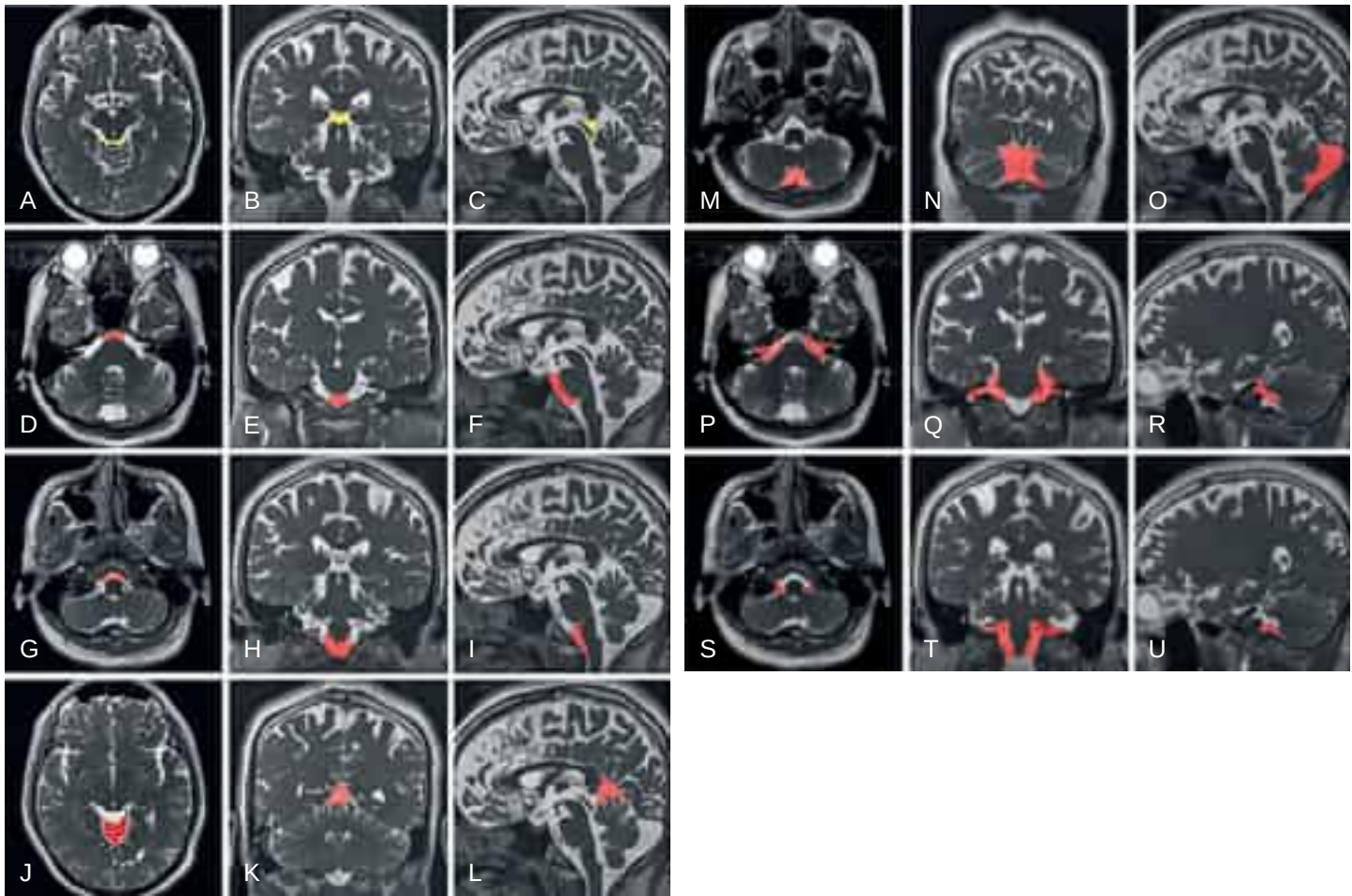


Fig. 18.17 Intratentorial subarachnoid cisterns. FIESTA MR images colour-coded to provide a visual approximation to the borders of the individually named intratentorial cisterns. These images are arrayed so that each row depicts a single cistern, in axial plane (column 1), coronal plane (column 2) and sagittal plane (column 3) images. This montage is intended to indicate the general location for each cistern within the sea of multiple confluent CSF spaces around the brain. **A–C**, Midline quadrigeminal plate (vein of Galen) cistern (yellow). **D–F**, Midline preponine cistern. **G–I**, Midline premedullary cistern. **J–L**, Midline superior vermian cistern. **M–O**, Midline cisterna magna. **P–R**, Paired cerebellopontine angle cisterns. **S–U**, Paired cerebellomedullary cisterns. (With permission from Rios JC, Galper MW, Naidich TP. Ventricles and intracranial subarachnoid spaces. In: Naidich TP, Castillo M, Cha S, Smirniotopoulos JG (eds), *Imaging of the Brain*. Chapter 13, 245–271. 2013, Elsevier, Saunders.)

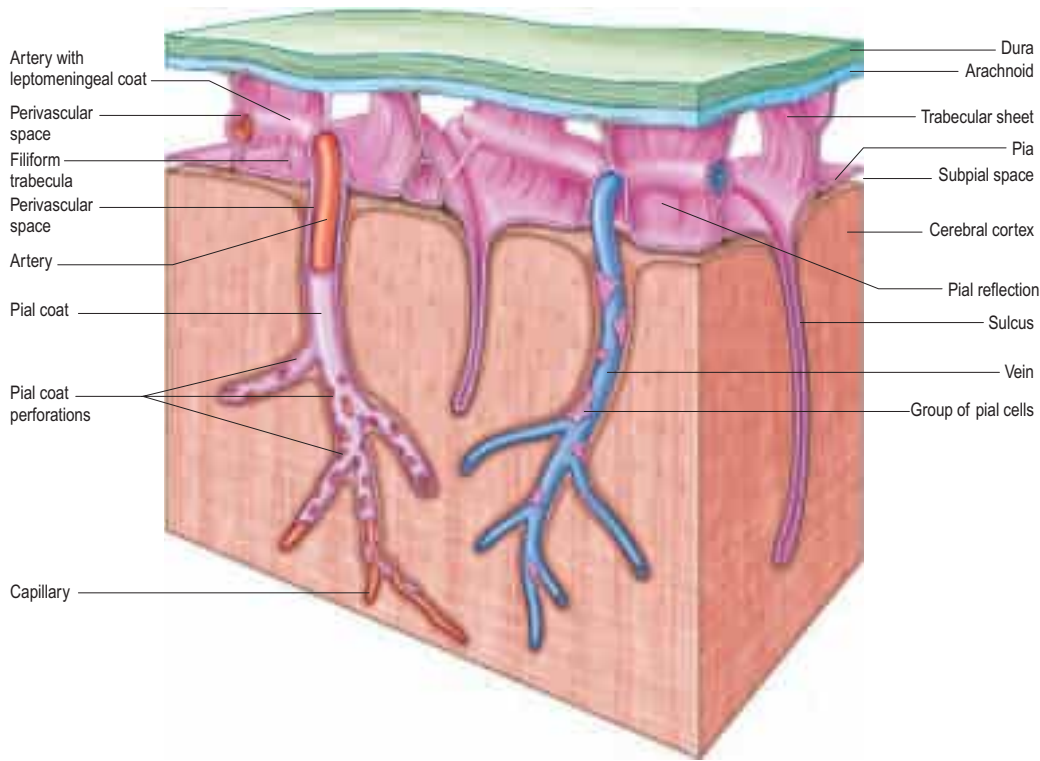


Fig. 18.14 The relationships between the dura mater, the leptomeninges and the blood vessels that enter and leave the cerebral cortex. The subarachnoid space is divided by trabeculae.

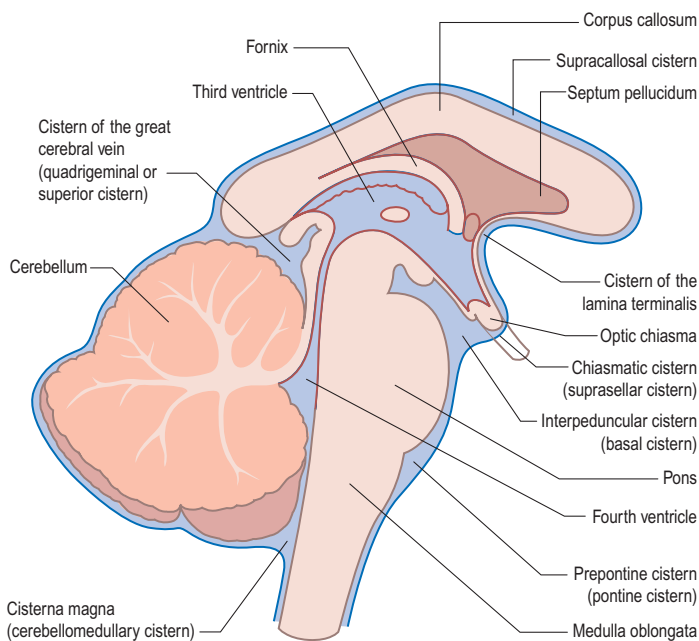


Fig. 18.15 The principal subarachnoid cisterns, sagittal section.

regions, thereby possibly facilitating directional flow of CSF through the space. A thin layer of leptomeninges, often only one cell thick, coats the trabeculae, vessels and nerves that cross the subarachnoid space; it fuses with the arachnoid mater at the margins of the exit foramina in the skull and vertebral column.

Arachnoid and pia mater are in close apposition over the convexities of the brain, such as the cortical gyri, whereas concavities are followed by the pia but spanned by the arachnoid. This arrangement produces a subarachnoid space of greatly variable depth that is location-dependent. The more expansive spaces form subarachnoid cisterns (Figs 18.15–18.17), which are continuous with the general subarachnoid space and crossed by long, filamentous trabeculae (Inoue et al 2009).

The largest cistern, the cisterna magna (cerebellomedullary cistern) is formed where the arachnoid bridges the interval between the medulla oblongata and the inferior surface of the cerebellum. The cistern is continuous above with the lumen of the fourth ventricle through its median aperture, the foramen of Magendie, and below with the spinal subarachnoid space. It contains the vertebral arteries and the origins of

the posterior inferior cerebellar arteries, the glossopharyngeal, vagus, accessory and hypoglossal nerves, and the choroid plexus.

On either side, the paired cerebellopontine cisterns (angle cisterns or cerebellopontine angle cisterns) are situated in the lateral angle between the cerebellum and the pons. They are traversed by the trigeminal, facial and vestibulocochlear nerves, the anterior inferior cerebellar arteries and the superior petrosal veins.

The prepontine cistern (pontine cistern) is an extensive space ventral to the pons, which is continuous below with the spinal subarachnoid space, behind and laterally with the cerebellopontine cisterns, and rostrally with the interpeduncular cistern. The basilar artery runs through the pontine cistern into the interpeduncular cistern, which also contains the origins of the anterior inferior cerebellar and superior cerebellar arteries, and the abducens nerves.

The quadrigeminal cistern (superior cistern, vein of Galen cistern, peripineal cistern) lies posterior to the brainstem and third ventricle, and occupies the interval between the splenium of the corpus callosum and the superior cerebellar surface. The great cerebral vein (formed by the union of the basal veins of Rosenthal and the internal cerebral veins) traverses this cistern and the pineal gland protrudes into it. It also contains the P3 segments of the posterior cerebral arteries and portions of the medial and lateral posterior choroidal arteries.

The paired ambient cisterns are extensions of the quadrigeminal cistern. On each side, they lie between the parahippocampal and dentate gyri of the temporal lobes laterally and the lateral surface of the midbrain behind the cerebral peduncles medially; they therefore extend laterally and anteriorly around the midbrain on either side to the interpeduncular cistern (basal cistern) (see Figs 18.16, 21.3). They are divided into supra- and infratentorial compartments and contain the basal vein of Rosenthal, the P2 segment of the posterior cerebral artery, the superior cerebellar artery and the trochlear nerves.

The average width of the ambient cistern in children is greatest at 1–3 years (2.8 ± 0.6 mm) and lowest at 4–10 years (2.4 ± 0.6 mm); it is reported to be larger anteriorly in boys (Skadorwa et al 2010).

The paired crural cisterns lie between the posterior borders of the unci and the cerebral peduncles on each side and contain the P1 segment of the posterior cerebral artery and the postoptic part of the anterior choroidal artery. The midline interpeduncular cistern (basal cistern) lies between the two cerebral peduncles, bordered superiorly and inferiorly by the diencephalic and mesencephalic leaves respectively of Liliequist's membrane. It contains the basilar artery, the origins of the posterior cerebral and thalamoperforating arteries, the proximal portion of the anterior pontomesencephalic vein and the proximal portions of the oculomotor nerves. Anteriorly, the interpeduncular cistern extends to the optic chiasma.

The paired cisterns of the lateral fossa (Sylvian cistern) are formed on either side by the arachnoid as it bridges the lateral sulcus between

the frontal, parietal and temporal opercula and contain the middle cerebral artery and veins.

Several smaller cisterns have been described. They include the chiasmatic cistern (suprasellar cistern), which is related to the optic chiasma, and the cistern of the lamina terminalis and the supracallosal cistern, all of which are extensions of the interpeduncular cistern and contain the anterior cerebral arteries. The subarachnoid space extends through the optic foramina into the orbits, where it is bounded by the sheath of the optic nerve. The latter is formed by fusion of the arachnoid mater and dura mater, and surrounds the orbital portion of each optic nerve as far as the back of the globe, where the dura fuses with the sclera of the eyeball. For further reading, see [Rios et al \(2013\)](#).

CIRCULATION OF CEREBROSPINAL FLUID

The total volume of CSF in the adult ranges from 140 to 270 ml. The volume of the ventricles is about 25 ml. CSF is produced at a rate of 0.2–0.7 ml per minute (600–700 ml per day), which means that the entire CSF volume is replaced approximately 4 times per day.

In children, CSF production, as measured by output through external ventricular drainage, increases during infancy, reaching 64% of the output of a 15-year-old child by 2 years of age. There is a logarithmic increase in CSF output with age and body weight ([Yasuda et al 2002](#)).

Mixing of CSF from different sources within the subarachnoid space is probably assisted by cilia on the ependymal cells lining the ventricles and by arterial pulsations.

Arachnoid villi and granulations

Arachnoid villi and granulations (Pacchionian bodies) are focal pouches of arachnoid mater and subarachnoid space that protrude through the walls of the dural venous sinuses, very often close to points where veins enter the sinuses. They are found in the superior sagittal, transverse, superior petrosal and straight sinuses in decreasing frequency and are most prominent along the margins of the great longitudinal fissure, commonly in the lateral lacunae of the superior sagittal sinus.

At the base of each arachnoid granulation, a thin neck of arachnoid mater projects through an aperture in the dural covering of a venous sinus and then expands to form a core of collagenous trabeculae and interwoven channels ([Fig. 18.18](#)). An apical cap of arachnoid cells, some 150 µm thick, surmounts the core and is attached to the endothelium of the sinus over an area some 300 µm in diameter. The rest of the core of the granulation is separated from the endothelium by a fibrous dural cupula. Channels extend through the cap to reach the subendothelial regions of the granulation. Meningiomas, the most common benign intracranial tumours, arise from arachnoid cap cells.

Dural depressions in the venous wall containing microscopic clusters of arachnoidal cells penetrating between the dural fibres are seen in the twenty-sixth fetal week. At 35 weeks they develop into arachnoid villi, which are simple protrusions through the dura. At full term, the number and complexity of the villi increase. It is generally believed that the fetal and early infantile villi do not fully penetrate the dural wall of the cranial venous sinus. With maturation they grow in size, number and structural complexity, achieve penetration through the wall of the venous sinus, and are thereafter referred to as arachnoid granulations. Granulations are visible to the naked eye in the parieto-occipital region of the superior sagittal sinus by the age of 18 months. In the elderly, the arachnoid membrane thickens and the arachnoid granulations degenerate and often calcify.

The function of arachnoid granulations is incompletely understood.

Hydrocephalus

Obstruction of the circulation of CSF leads to accumulation of fluid within the ventricular system (hydrocephalus; [Fig. 18.19](#)). Within the brain, critical points at which obstruction may occur correspond to the narrow foramina and passages of the ventricular system. Thus, obstruction of the interventricular foramen causes enlargement of the lateral ventricles; obstruction of the cerebral aqueduct leads to enlargement of both the lateral ventricles and the third ventricle; and obstruction or congenital absence of the apertures of the fourth ventricle leads to enlargement of the entire ventricular system. Membranes in the subarachnoid cisterns can cause extraventricular obstruction to CSF flow. Hydrocephalus in the presence of intact communications between the ventricular and subarachnoid spaces, 'communicating hydrocephalus', is most often associated with defective absorption of CSF.

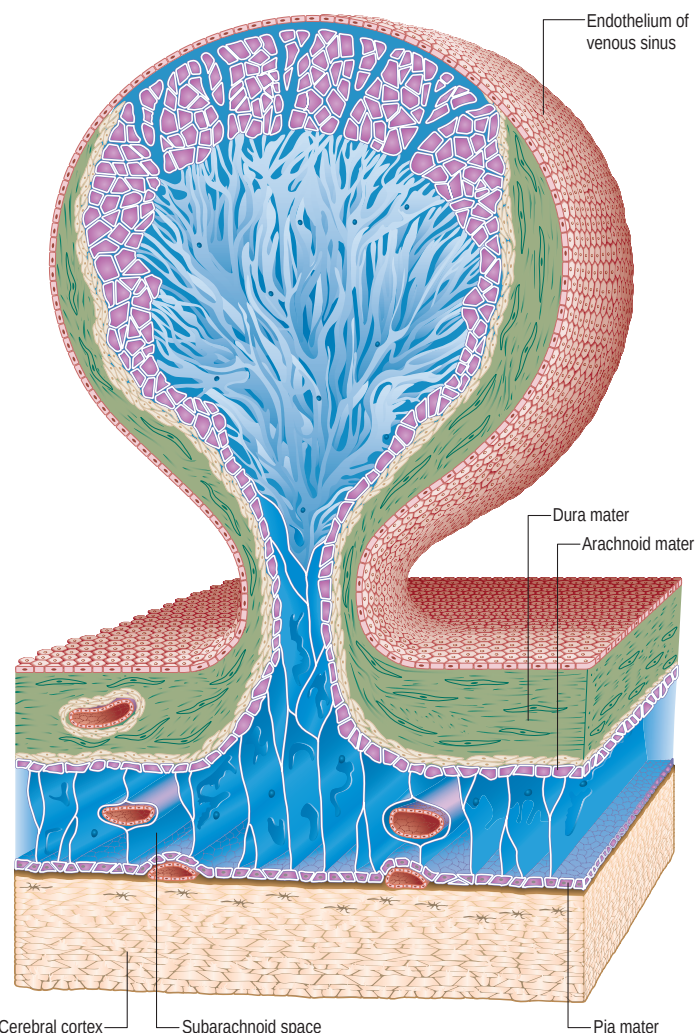


Fig. 18.18 An arachnoid granulation. (Modified with permission from Springer-Verlag, *Principles of Pediatric Neurosurgery*, Vol 4: Morphology of CSF drainage pathways in man. Raimondi (ed), [Kida and Weller 1994](#).)

PIA MATER

The pia mater is a delicate membrane that closely invests the surface of the brain and spinal cord, from which it is separated by a microscopic subpial space. It follows the contours of the brain into concavities and the depths of fissures and sulci (see [Fig. 18.14](#)). In the spinal canal, it contributes to the formation of the dentate ligaments and the filum terminale ([Adeeb et al 2013](#)).

Pia mater is formed from a layer of leptomeningeal cells, often only 1–2 cells thick, in which the cells are joined by desmosomes and gap junctions but few, if any, tight junctions. The cells are continuous with the coating of the subarachnoid trabeculae and separated from the basal lamina of the glia limitans by bundles of collagen, fibroblast-like cells and microscopic vessels that lie in the subpial space ([Mercier et al 2002](#)) (see [Fig. 18.14](#)).

It was long thought that the subarachnoid space was connected directly with the perivascular spaces (Virchow–Robin spaces) that surround blood vessels in the brain. However, it is now recognized that the pia mater is reflected from the surface of the brain on to the surface of blood vessels in the subarachnoid space, which means that the subarachnoid space is separated by a layer of pia from the subpial and perivascular spaces of the brain (see [Fig. 18.14](#)). The pia mater appears to form a regulatory interface between the subarachnoid space and the brain, across which small molecules may diffuse between the interstitial space and the CSF through the perivascular space. This may allow movement of metabolites and could play a role in blood–brain barrier function ([Zhang et al 1990](#)).

During development the pia mater becomes apposed to the ependyma in the roof of the telencephalon and fourth ventricle to form the tela choroidea. It shares a common embryological origin and structural similarity with the arachnoid mater.

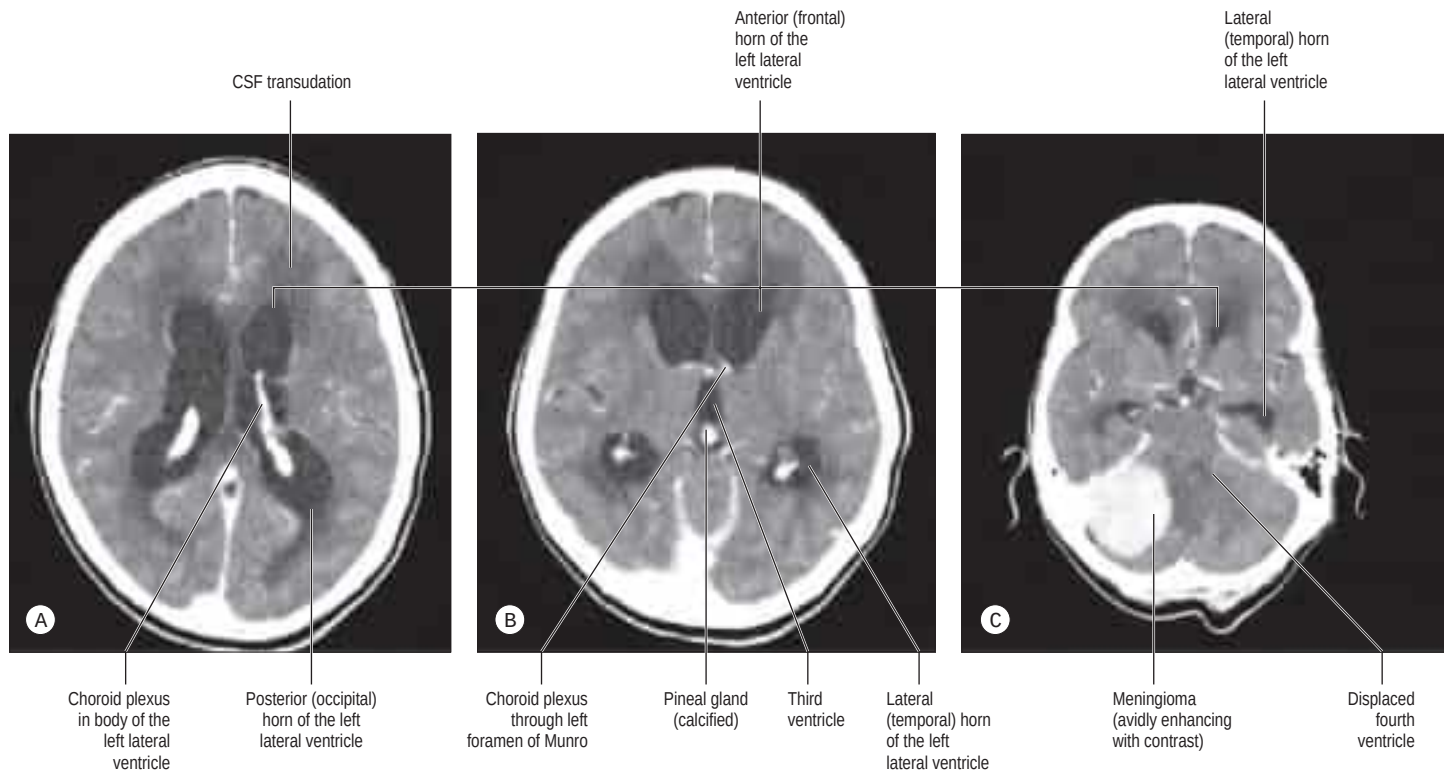


Fig. 18.19 Axial CT images post contrast showing acute hydrocephalus (from posterior fossa meningioma). Note the transudation of fluid into the juxtaventricular white matter secondary to acute hydrocephalus.

Magnetic resonance imaging with fluid movement-sensitive sequences performed on normal subjects has shown CSF flow to be pulsating and bidirectional between the compartments of the ventricular system, probably with a net flow in a direction towards the fourth ventricle (Yamada 2008). The flow of CSF changes at the age of 2 years from an infantile to a mature pattern. The infantile pattern is characterized by flow directed into the ventricular system, and the mature pattern by flow directed out of the ventricles (Bateman and Brown 2012). The route of CSF through the subarachnoid space is even more complex and less understood, and it is debated whether CSF moves mainly over the cerebral convexities or mainly at the cranial base. It is known that CSF is absorbed into the superior sagittal sinus, but also that this site is insufficient to account for all CSF absorption and that this absorption system may only start to operate when CSF pressure is increased. Therefore one or more additional routes of absorption must exist but their precise location, function and mutual contributions to CSF absorption have yet to be clarified (Pollay 2010). Absorption through pial and subarachnoid vessels, across the ependymal ventricular surfaces and pial surfaces and via extracranial lymphatics may all be involved (Zakharov 2004).

It has long been thought that arachnoid villi and arachnoid granulations constitute the major pathway for the absorption of CSF from the subarachnoid space into the blood, and there are observations both to support and to question this. Their relative paucity in the fetal and early postnatal period strongly suggests the importance of other absorption routes. The degenerative changes in arachnoid granulations in old age could be the substrate for the condition of 'normal pressure hydrocephalus' and for decreased CSF absorption, which could play an important role in degenerative brain diseases of the elderly.

An alternative, and not necessarily mutually exclusive, view of arachnoid granulations is that they are involved in dissipating the pressure wave that occurs in the subarachnoid space during arterial systole (caused by the pulsation of the arterial vessels within the space). Since the dural venous sinuses lie within thick, non-compliant dural coverings, pressure changes within them functionally represent the extracranial venous compartment; systolic dilation of the arachnoid granulations would therefore dissipate intracranial pressure changes into the extracranial venous system.



Bonus e-book images and videos

Fig. 18.2 The ventricular system.

Fig. 18.16 Supratentorial subarachnoid cisterns.

Fig. 18.17 Infratentorial subarachnoid cisterns.

Fig. 18.19 Axial CT images post contrast showing acute hydrocephalus (from a posterior fossa meningioma).

Video 18.1 Interactive 3D rotation of the subarachnoid space.

Video 18.2 Interactive 3D rotation of the ventricles and cisterns.

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